

Algorithm showcase:

goal: easier to catch the idea
 easy to check if the algo is correct.

Idea:



$$L(h) = \left\{ v : \forall w \in V(h) \text{ and } L(w) = L(v-1) \right. \\ \left. (w, v) \notin E(h), L(v) = 0 \right\}$$

$$\text{Protect}(L) = \left\{ v : \exists w \in V(h) \text{ and } L(w) = L(v-1) \right. \\ \left. \text{and } (w, v) \in E(h) \right. \\ \left. L(v) = L_1 \right\}$$

$$\text{Protect}(L) = \left\{ \text{stable} : \text{die} \right\}$$

Eq Class (i,j)
 (ornik of
 (i,j))

$$\text{Eq Class (i,j)} \\ = \text{Eq Class (i,j)} - \text{Eq Class (i,j)}$$

• loop over l -levels:

• loop over n in l :

• loop over p protect $(l+1, n)$:

• loop over w warn $(l+1)$:

if first:

$$eq_{temp} = Eq[l, p, w]$$

eq temp, all element of $Eq(l, p, w)$

$$Eq[l, p, w] = eq_{temp}.$$

unify the eq. classes for all p .

(1) fill warnings and protected

(2) for $l = 1, 2, \dots, L(h)$

a) concatenate Eq in class of all the
unification of each protected list of l

b) concatenate Eq in class of warnings

to ~~with~~ ~~set~~ of protocols



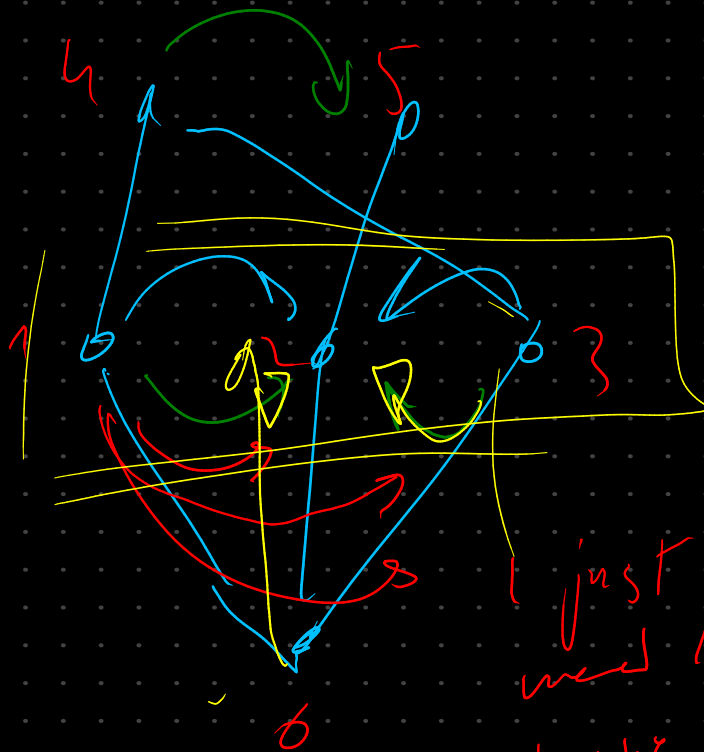
$$E_{12} = E_{23} = E_{13}$$

$$E_{31} = E_{12} = E_{23}$$

$$E(1,2) = (4, 1)$$

$$E(1,3) = (5, 4)$$

$$E(3,1) = (4, 5)$$



I just need to prove that

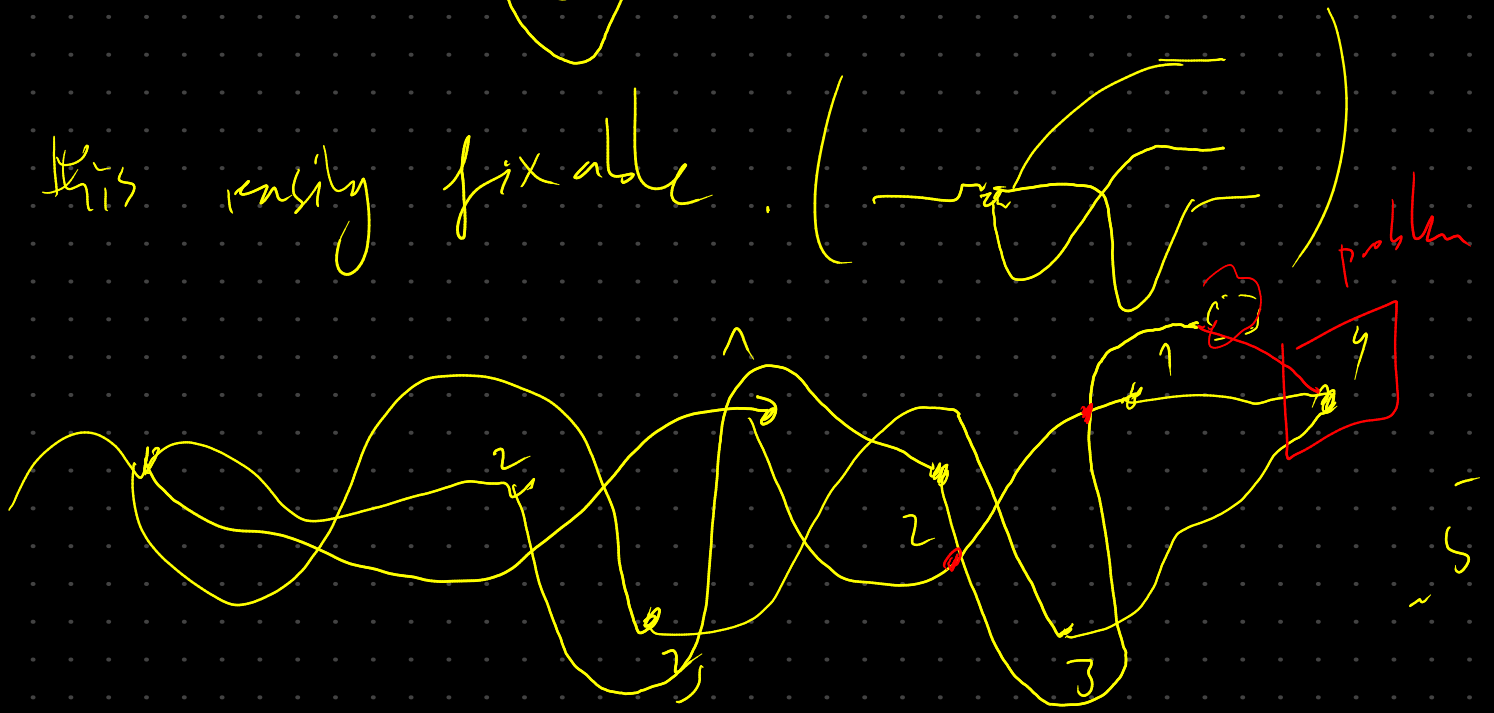
there are not a issue

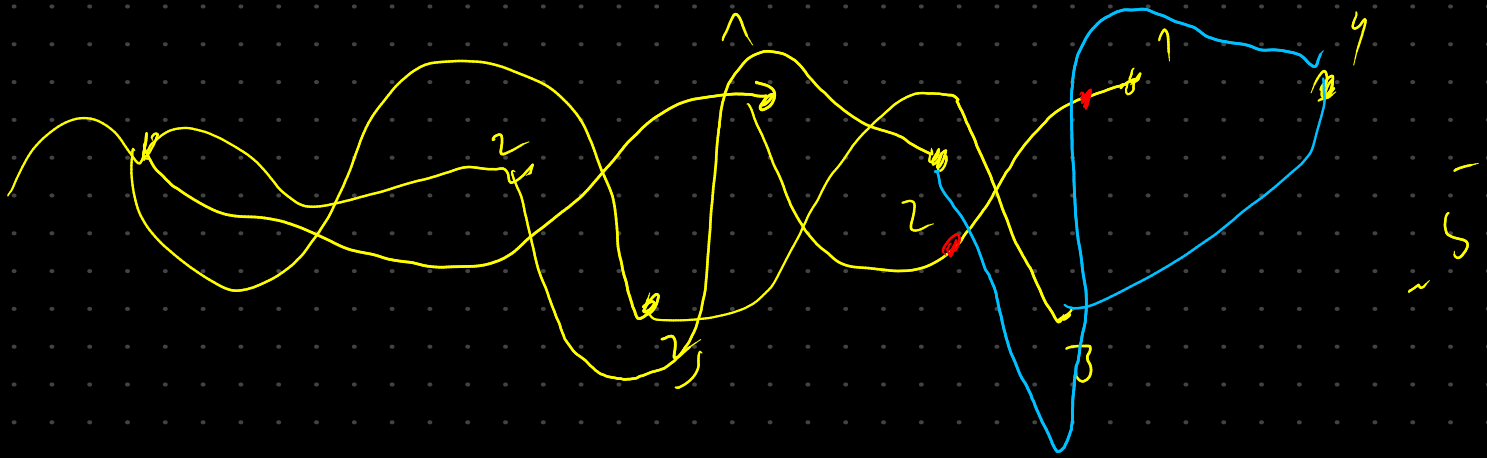
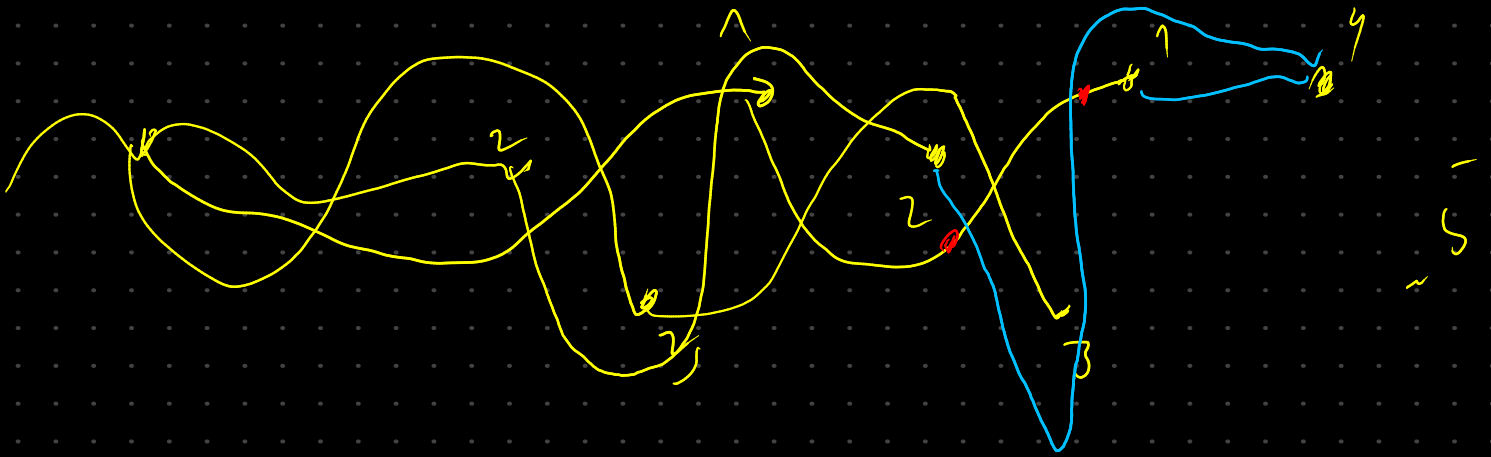
Can I prove that we just need to
check the vertices that have no
planarity constraint

$$(1, 2) - (3, 2)$$



this easily fixable. ()

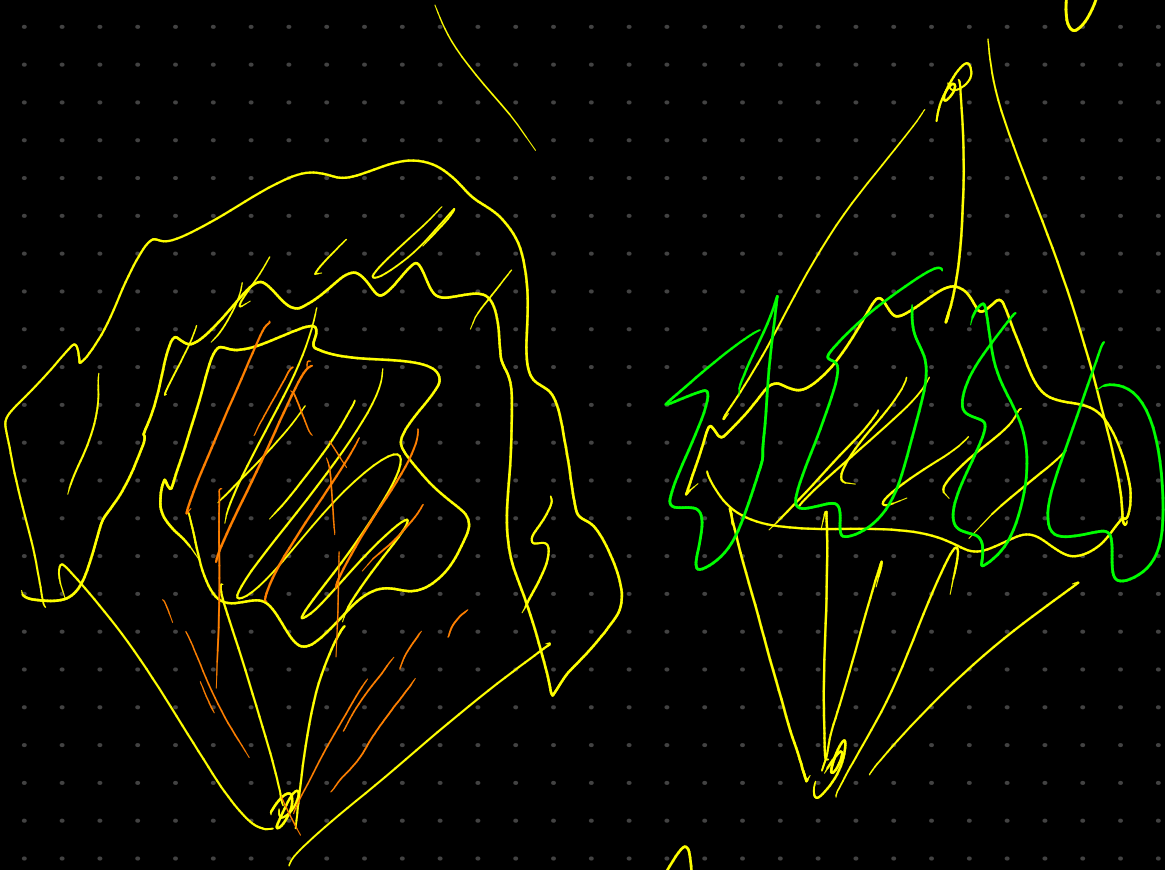




basically we get in this situation
when either all the submaps are disjoint
or there is a shortest cycle between two
and the third ends before reaching the cycle.

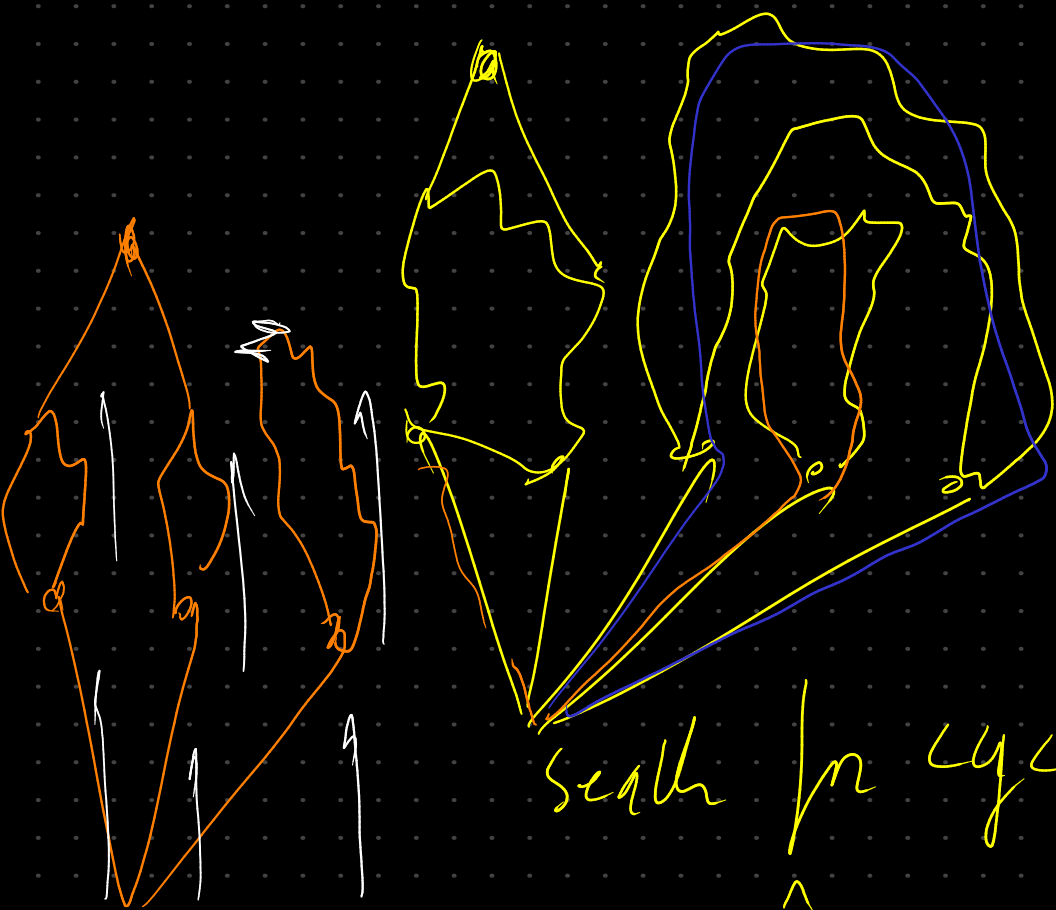
How can we discover these adj.
edges?

- Can we work with all adj. edges?

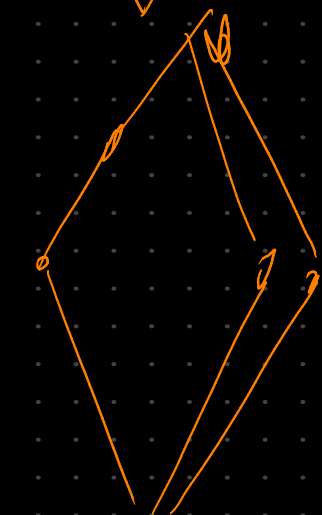
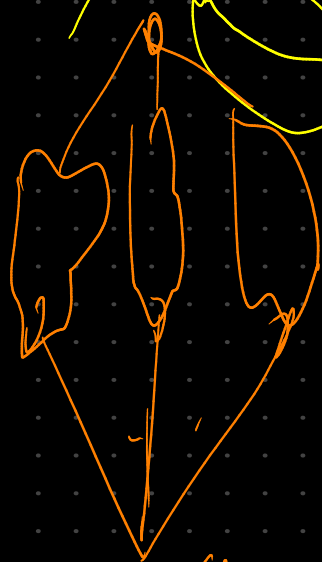


Can we do
whatever

We can
have all
of the
cases

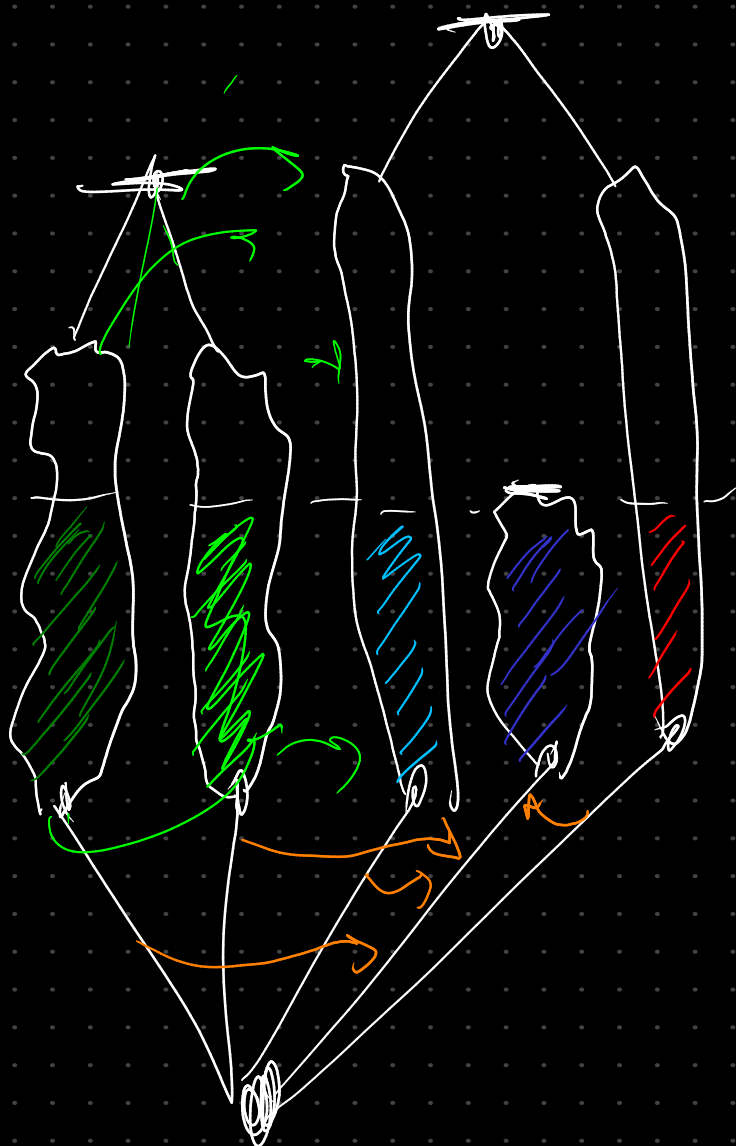
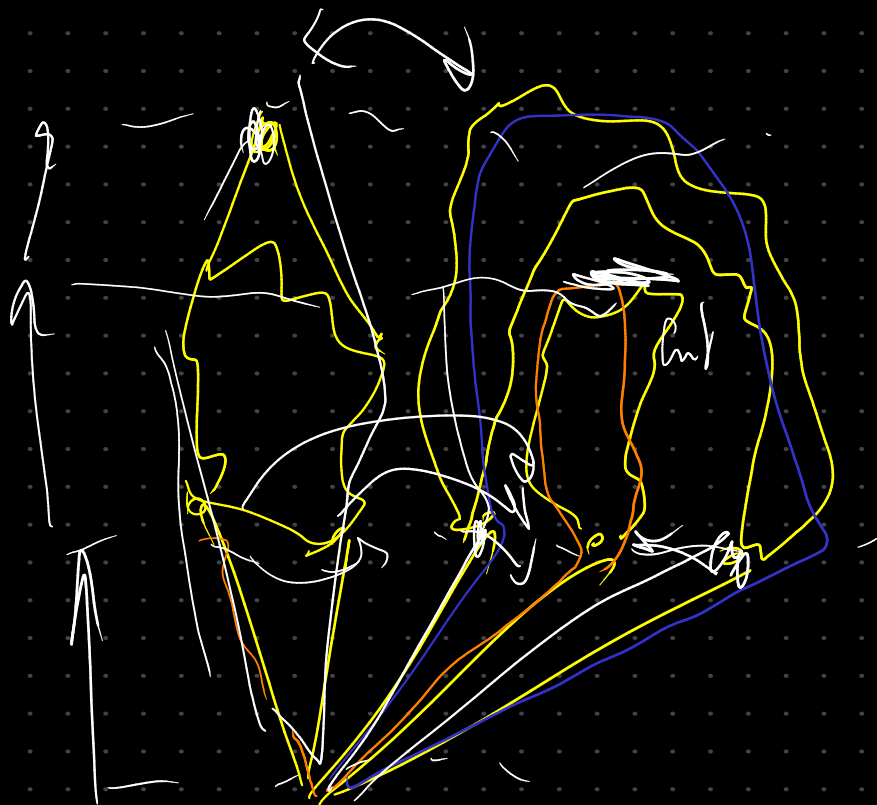


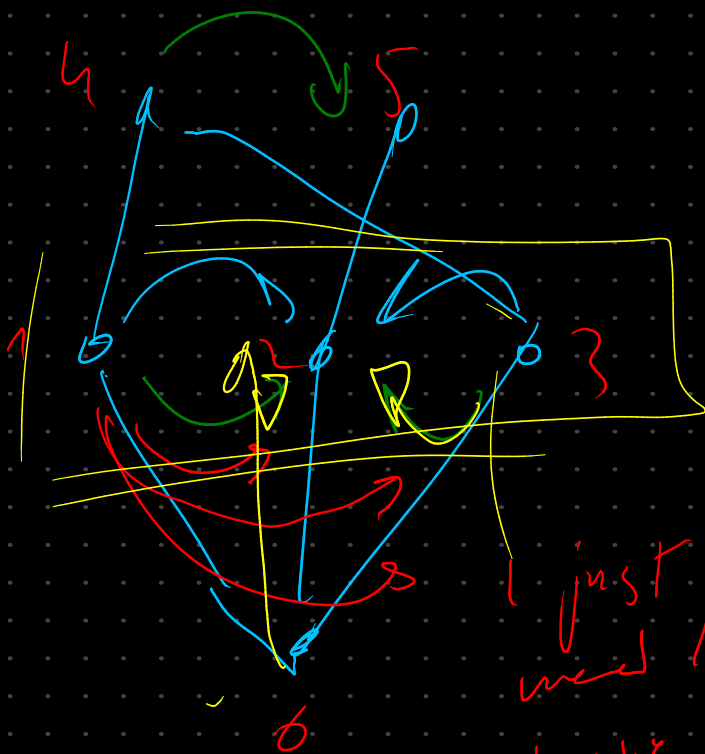
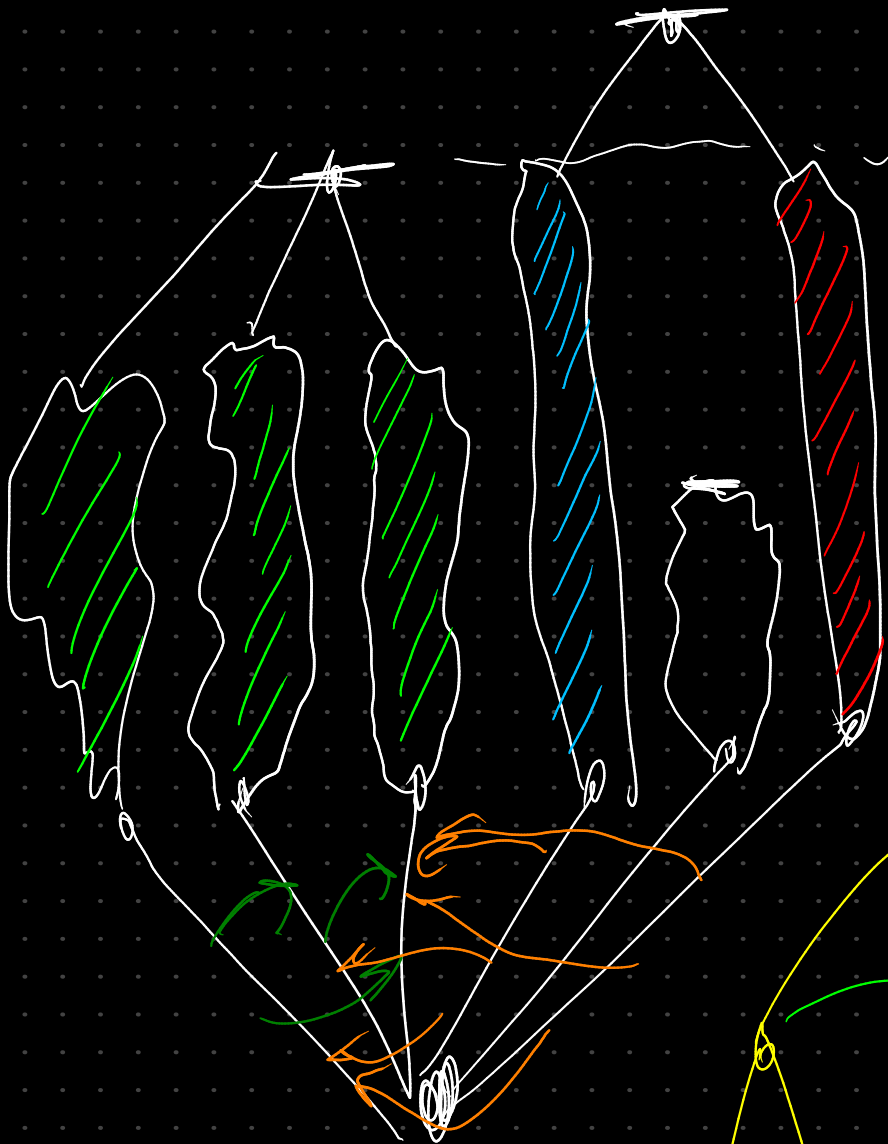
seath in cycles



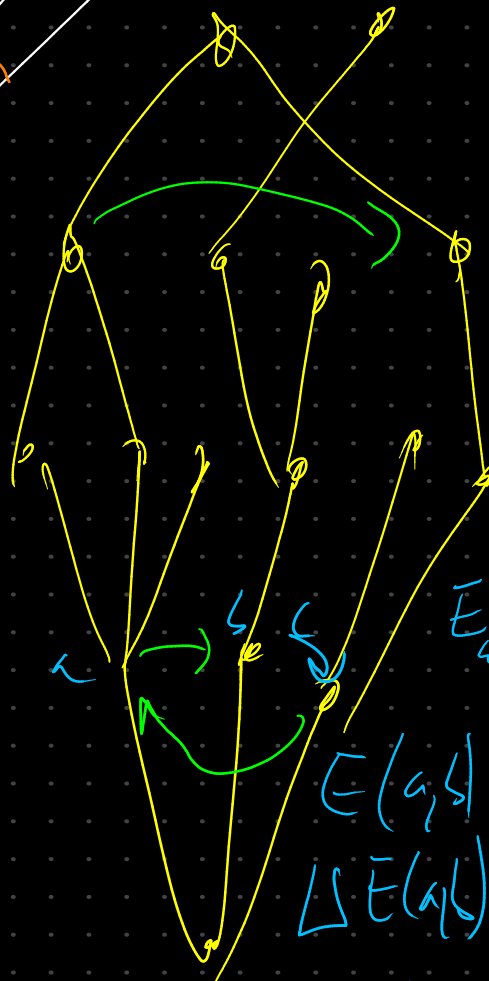
check camps
with $h \sim \{v_0\}$







I just need to prove!

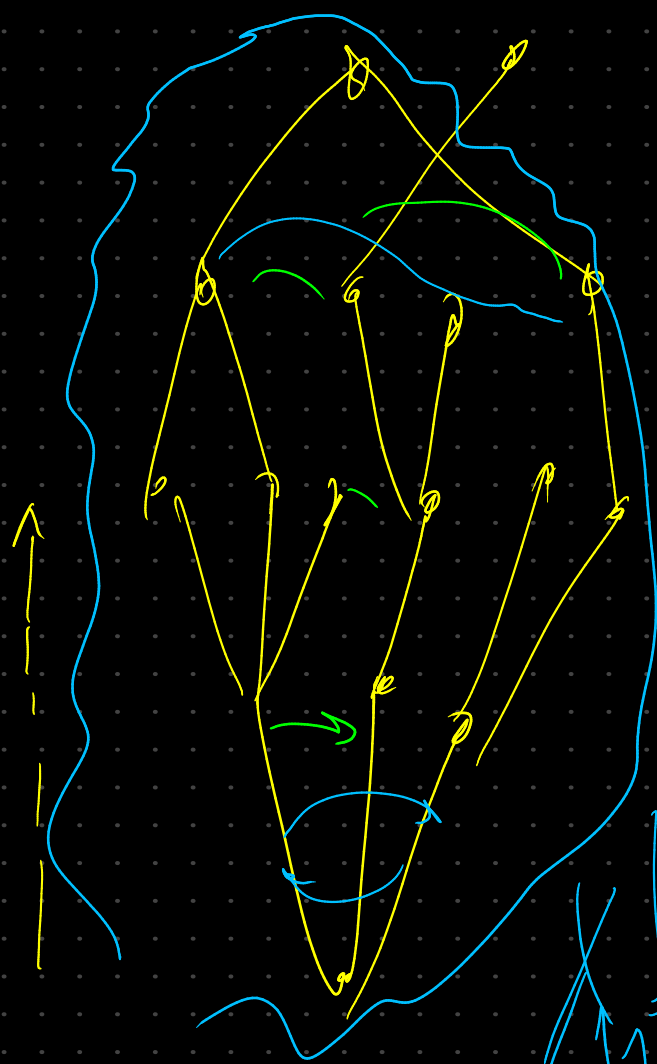


$$\bar{E}_{\text{ass}}(a, b) = \bar{E}(a, b)$$

$$\begin{aligned} &E(a, b) \cap \bar{E}(a, b) \\ &\subseteq \bar{E}(a, b) \cap \bar{E}(c, a) \\ &\neq \emptyset \end{aligned}$$

$$\underline{E_g(a, b)} = \{K, a\}$$

$$E_g(a, b) = \bar{E}_g(b, a)$$



$$(a,b) \in E_1$$

$$\{(a,b), (c,d)\}$$

$$E_1 = \{(a,b), (c,d)\}$$

we map first the
point that we relate

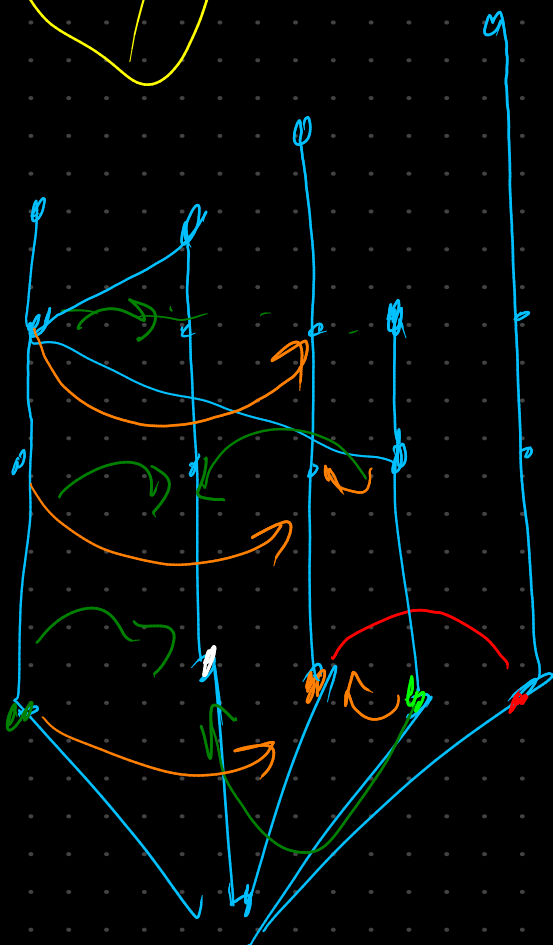
$$X_1 = \{(a,b), (c,d)\}$$

$$X_2 = \{(c,d)\}$$

suppose we have

$$x_1, \dots, x_n$$

$$x_1, x_2, x_3, \dots, x_n$$



I just have to prove:

- we don't remove any relation from the eq. classes $\pm$ (planarity nodes)
- for the three cases crossing non cyclic relation between triplet of vertices work.